Sunil Somarajan

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EDUCATION AND TRAINING

05/10/2021 – 07/05/2024 Kollam, India

BACHELOR IN COMPUTER APPLICATION Fatima Mata National College (Autonomous)

Website https://fmnc.ac.in/ | **Field of study** Computer Applications | **Final grade** 7.24 | **Level in EQF** EQF level 6

2019 – 2021 KOTTARAKKARA , India

HIGHER SECONDARY EDUCATION EVHSS NEDUVATHOOR

Website https://schools.org.in/kollam/32130700601/evhss-neduvathoor.html | **Field of study** Biology Science | **Final grade** 84% |

Level in EQF EQF level 4

2018 – 2019 Kottarakkara, India

SECONDARY SCHOOL EDUCATION Sree Narayana Guru Central School

Website https://www.sngcs.org/ | **Field of study** Generic programmes and qualifications | **Final grade** 80% |

Level in EQF EQF level 3

PROJECTS

01/2024 – 04/2024

SIGN SENSE

Objective: Sign language serves as a crucial means of communication for individuals with hearing impairments, offering a visual language that relies on hand gestures, facial expressions, and body movements. This project aims to develop a Sign Language Recognition Application using Python leveraging machine learning techniques to interpret and translate sign language gestures into meaningful text or spoken language.

Technology Stack:
Python
JAVA
IDE (PyCharm,Android Studio)
Cv2 (OpenCV)
TensorFlowlite
Firebase

Role: Full-stack Developer

Key Tasks:

1. Developed a machine learning model using TensorFlow and OpenCV for real-time sign language gesture recognition and integrated it with TensorFlow Lite for on-device inference.

2. Designed and developed the frontend using Android Studio and Java, creating a user-friendly interface for capturing and translating hand gestures into text or spoken language.

3. Implemented real-time gesture detection with OpenCV, using image preprocessing techniques to improve recognition accuracy.

4. Integrated Firebase for user data management, including storing recognized gestures and maintaining translation history.

5. Tested and optimized the application, improving model accuracy and responsiveness across Android devices.

6. Deployed the application, gathered user feedback, and refined the system for better accuracy and user experience.

02/2023 – 06/2023

TRAFFIC SAFETY GUIDE WEB APPLICATION

Objective: The AI Camera Alerter and Speed Alerter website is a web application that utilizes HTML, CSS, and JavaScript along with the Geolocation API and Leaflet Map API. The primary objective of this project is to provide real-time alerts and notifications to users based on their location and speed. The website incorporates AI-powered camera detection to identify specific locations of interest. When a user enters a designated area, the website triggers an alert, notifying the user of their presence in the defined zone. The alerts are displayed using visually appealing HTML and CSS designs.

Technology Stack:
Html
Css
JavaScript
Microsoft Edge

Role: Full-stack Developer

Key Tasks:

1. Developed the frontend using HTML, CSS, and JavaScript to create a responsive user interface with real-time alerts and notifications.

2. Integrated the Geolocation API to track users' location and dynamically trigger alerts when entering predefined zones.

3. Implemented the Leaflet Map API for displaying real-time maps and marking specific zones of interest the users.

4. Built AI-powered camera detection features to identify and notify users of their presence in designated areas.

5. Designed visually appealing alerts using HTML and CSS to ensure clear and effective communication of notifications.

6. Tested cross-browser compatibility, ensuring the website functioned smoothly on Microsoft Edge and other browsers.

CONFERENCES AND SEMINARS

07/2023 – 12/2023 Fatima Mata National College

Brain Gate Technology

Brain gate is a neuroprosthetic device that converts brain activity into computer commands.It is an electrode chip implemented in the brain which contains an internal signal sensor and external processor that converts neural signals into an output signal.

DIGITAL SKILLS

Programming Languages: C, Python, PHP | Software Development: Java, C++ | Database Management: SQL, DBMS | Web Technologies: HTML, CSS, Javascript, PHP | Microsoft Office package: Microsoft Word, Excel, PowerPoint, Access | 3D Animation | Adobe Photoshop (basic elements) | Google Suite (Doc, Slides, Form, Sheet, Drive) | Video Conferencing (Zoom, Teams, Skype, Webex) - Advanced | Browsing the Internet (experienced) | Social Media including Facebook , WhatsApp and Twitter

IELTS

Overall Score: 6.5

Listening: 7.5, Reading: 6.5, Writing: 6, Speaking: 6.5

LANGUAGE SKILLS

Mother tongue(s): **MALAYALAM**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	B2	B2	C1	B2
TAMIL	B2	A1	B2	B2	A1
HINDI	B2	B2	B2	B2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HONOURS AND AWARDS

23/07/2024

Bachelor's Degree – Fatima Mata National College

18/07/2019

3D Animation – Kerala State Rutronix

HOBBIES AND INTERESTS

Coding, Gamming, Social Media, Movies, Reading,